Kurt **Waage**

Japan

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Summary

Gameplay programmer in Unreal Engine, C++, and Blueprint, with live-service and early-development experience. Strengths include designer-friendly systems, testable code, clear technical documentation, and debugging. Brings a systems-integration background from robotics and autonomous vehicles to building reliable gameplay.

Skills

C++, C#, Unreal Engine, Blueprint, Replication, GAS, UMG & Common UI, MVVM, Control Rig, Enhanced Input, Chaos Physics, Perforce, SVN, Rider, Visual Studio, Jira

Gameplay Systems, Prototyping, Multiplayer, Version Control, Debugging, Optimization, Consoles (Xbox & PlayStation)

Work Experience

Stage Games — *Japan*

Gameplay Programmer — *Jan. 2026 - Present*

- Building core gameplay systems from prototype toward production, in Unreal Engine 5 with C++ and Blueprint
  - Architected a networked interaction framework where clients scan and rank targets locally while the server re-validates each request.
  - Built gameplay UI on Common UI and MVVM so C++ handles behavior and data flow while designers keep full control of layout and presentation.
  - Developed a data-driven building and placement system on voxel terrain, with a shared inventory moving resources between structures and vehicles.
  - Integrated the Gameplay Ability System to drive payer interaction and status, with configurable ability sets, and a gateway ability that routes interaction requests carrying custom target-data payloads.
  - Programmed gameplay-driven animation for Chaos Vehicles with Control Rig and Animation Blueprints, including FK crane articulation and hook and cable orientation solved at runtime.
  - Structured gameplay logic to be unit-testable without loading a level, backed by a dedicated automation module with unit and in-world functional tests.

Black Tower Studios — *Japan*

Game Programmer — *Aug. 2024 - Nov. 2025*

- Supported, maintained, and created new content for the live-service game The Texas Chainsaw Massacre using Unreal Engine 4.
  - Updated the interaction system for more natural player experience.
  - Created automated tests using Gauntlet for Windows, Xbox, and PlayStation, reducing release bugs.
  - Reworked and developed UI menu flows for game modes Rush Week and Custom Matches.
  - Optimized gameplay, animation, and network systems across multiple platforms.

Woven by Toyota — *Tokyo, Japan*

Research Engineer — *May 2022 - Nov. 2023*

- Researched, developed, and integrated software and hardware XR capability for selected autonomous vehicles
  - Developed APIs to connect autonomous systems with external applications.
  - Designed hardware fixtures and integrated external PCs and XR components.
- Built software and hardware triggers for emergency stop functionality, improving safety response times.
  - Provided on-site field support and conducted thorough post-test bug reviews.

Toyota Research Institute — *Cambridge, MA | Tokyo, Japan*

Systems Integration Engineer — *Oct. 2018 - May 2022*

- Co-led two vehicle fleet builds, completing 19 autonomous vehicles on schedule for the 2020 Olympics.
- Documented and reported technical issues, led bug scrubbing, and implemented verified fixes.
- Trained team members on vehicle builds, retrofits, maintenance, and troubleshooting.

Northeastern University — *Boston, MA*

Research Assistant, Robotics and Intelligent Vehicles Research (RIVeR) Lab — *Nov. 2016 - Jan. 2018*

- Conducted research in underwater robotics and monocular optical flow-based navigation and perception.

Orbital ATK — *Layton, UT*

Manufacturing Engineer — *Aug. 2015 - Apr. 2016*

- Supported aerospace manufacturing processes and quality assurance.

US Army — *Germany*

Sergeant – Infantry Section Leader — *Sept. 2005 - Feb. 2010*

- Honorably discharged from the US Army Infantry as Sergeant (E-5) with two deployments to Iraq.
- Taught Military Operations in Urban Terrain (MOUT) to foreign and domestic groups of platoon size or larger.

Education

Northeastern University — *Boston, MA*

M.Sc. in Mechanical Engineering — *2016 - 2018*

- Gordon Institute of Engineering Leadership fellow
- Concentration in Mechatronics

Colorado State University — *Fort Collins, CO*

B.S. in Mechanical Engineering and Engineering Science — *2011 - 2015*

- Minor in Mathematics
- Space Concentration